

HARRISON B. GOLDBERG

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EDUCATION

Arizona State University, Tempe, AZ

Expected May 2029

B.S. Mechanical Engineering | GPA: 3.37/4.00

- **Relevant Coursework:** MATLAB Programming, Linear Algebra, Calculus I–II, University Physics I: Mechanics
- **In Progress Coursework (Fall 2026):** Mechanics of Particles & Rigid Bodies I, Computer-Aided Engineering I, Calculus for Engineers III, Modern Differential Equations, University Physics II: Electricity & Magnetism, University Physics Lab II

Technion – Israel Institute of Technology, Haifa, Israel

2024 – 2025

Visiting Engineering Coursework

ENGINEERING PROJECTS

Bitzy (KineArm V1) – Five-Axis Modular Robotic Arm

June 2026 – Present

Personal Project | Designed & built hardware; AI-assisted software controls

- Owned mechanical design and fabrication of a five-axis, six-servo arm with paired shoulder actuators; iterated Fusion CAD, printed structures, bearing supports and servo fits through rapid prototyping and assembly.
- Built two interchangeable toolheads, a gripper and screwdriver, around a common mounting interface; refined gearing, spindle support and service access in CAD. Modular architecture supports an optional sixth axis.
- Integrated ESP32-S3/PCA9685 hardware with AI-assisted BLE and teach/replay controls; identified a 5.6° initial shoulder zero-reference error during bench calibration and used physical-datum and end-stop observations to revise calibration and command limits.

Acuivect – Precision Insertion Demonstrator

2026 – Present

Independent Project | CAD complete; physical build pending

- Completed AI-assisted Fusion CAD and fabrication drawings for an XYZ demonstrator with dual-camera mounts, removable probe cartridge, docking latch and moving-table brain-phantom fixture.
- Completed digital motion/interference and head structural-analysis checks; developed a browser simulation of docking, insertion and replay through AI-assisted workflows. Physical accuracy remains untested.

AUREX H20 – Tendon Robotic Hand

2026 – Present

Personal Project | Mechanical CAD complete; actuation revision in progress

- Completed AI-assisted CAD for a 20-joint hand with removable covers and TPU pads; checked constraints, poses and print meshes. Eight-servo revision underway; fabrication, full tendon routing and controls validation pending.

Eight-Servo Quadruped Robot

2026 – Present

Personal Project | Built hardware; AI-assisted firmware

- Built an ESP32/PCA9685 quadruped; calibrated two joints per leg for stable standing and repeatable Xbox-controlled gait. AI-assisted firmware includes four staggered gait phases, per-leg settings and MPU6050 pitch/roll compensation.

PROFESSIONAL EXPERIENCE

TinkerTank – Founder & Product Designer

2025 – Present

- Design, prototype, manufacture and sell PETG aquarium components on Etsy; translate customer needs and equipment constraints into parametric CAD, iterate physical fit, and fulfill orders.

Harbegold LLC / Fino – Founder

2026 – Present

- Launched Fino, a financial-learning iOS app, on the App Store through AI-assisted development.

Engineering Design Club – The Heschel School, NYC

2022 – 2024

- Founded and co-led the club; taught Tinkercad, Fusion 360, 3D printing and additive-manufacturing workflows.

TECHNICAL SKILLS

CAD & Hardware: Fusion 360, SolidWorks, MATLAB, ESP32, Arduino, I²C, FDM printing, laser cutting, soldering

AI-Assisted Workflows: CAD and Python/C++ workflows using Claude Code and Codex

Languages: English (native), Hebrew (conversational)